Transfer Learning for Extreme Domain Gap

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Abstract

One of the most significant drawbacks of deep learning is that it requires a lot of training data. This problem makes it difficult to apply deep learning models to specialized fields, such as medical or industrial, where it is not easy to obtain sufficient data. The most common way to deal with this problem is finetuning a model, pretrained on general and large-scale datasets like ImageNet, with the target data. However, when the gap between the source and target domains is extremely large, this pretrain-finetune framework may not be of much help. In this study, we propose a simple but strong method for transfer learning when the domain gap between the source and target domain is extremely large. Specifically, we design cross-domain mixup-noise (CDMN) to augment a small amount of labeled target data, and a cross-domain joint supervised learning method to effectively leverage augmented target training samples. We achieve state-of-the-art performance on various benchmarks and systemically analyze factors contributing to transfer learning performance through extensive ablation studies.

1. Introduction

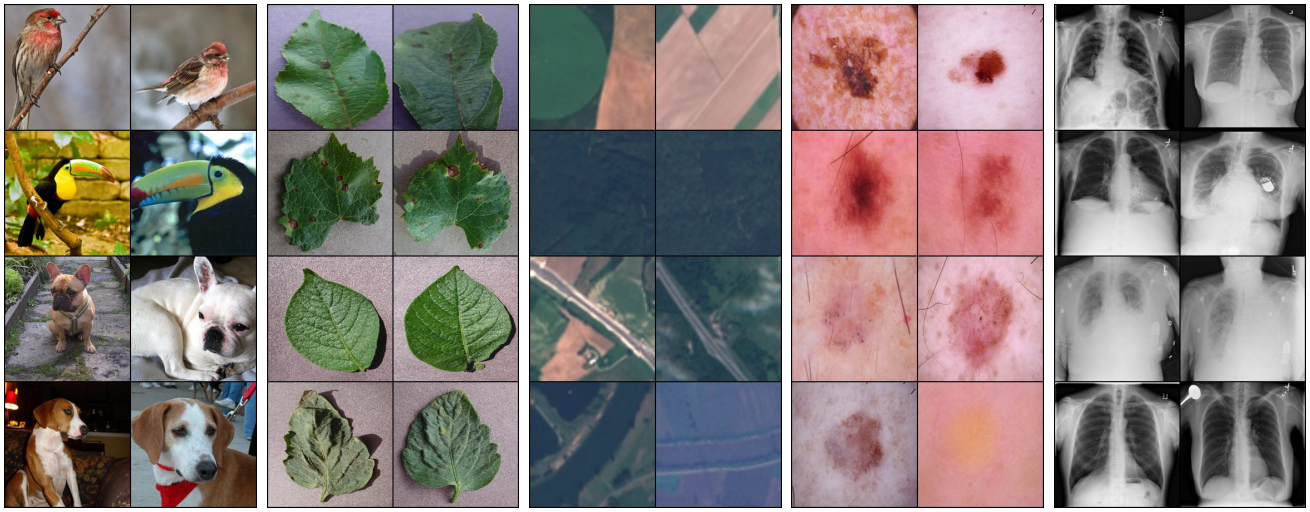
Although deep learning models have succeeded in many fields, they requires much training data for good performance. However, it is not always possible to obtain sufficient data in the target domain. In particular, it is more difficult in fields like medical imaging, where only experts can annotate. To overcome this limitation, it is a standard process to pretrain the model on a general and large-scale source dataset such as ImageNet [6] and then finetune it on a target dataset. This method initializes the model with pretrained weights and then finetune that model into the target-specific model. This pretrain-finetune method facilitates the convergence of training and provides higher performance than learning from scratch with random weights.

To achieve better performance than simple finetuning, previous methods propose to selectively use source samples similar to the target samples [9, 5] or preserve pretrained source knowledge [44, 25]. These methods may be useful when the domain gap between the source and target domains is not significant (e.g., from natural source domain to natural target domain.) or the target classes are a subset of the source classes (e.g., from a coarse-grained source domain to fine-grained target domain.)

However, when the domain gap between the source and target domains is extremely large, features useful for classification in the source domain will differ from those for the target domain. For example, features for classifying dog breeds will be different from classifying chest X-ray images as shown in Figure 1. Therefore, previous methods are difficult to apply or rather degrade performance in these situations [3]. Especially in the medical domain, where samples can be annotated only with professional knowledge, general source knowledge will not be helpful, and only the precise target label can be used to learn the decision boundary. Even though this situation is prevalent and practical, little research has been conducted. Although domain adaptation [39] is a way to reduce the large domain gap between different domains and increase performance in the target domain, domain adaptation assumes a fully or partially overlapped label space, limiting its application. Accordingly, scenarios, where a similar source dataset or task is unavailable, have not been handled properly. In this paper, we propose a method to improve transfer learning performance when the target domain has very low similarity and disjoint label space compared to the source domain.

A recent study shows that using samples from the data manifold as noise helps to learn better representations than data-independent noise [36]. [23] shows that two domains that may look entirely different to humans can share features and proposes to utilize out-of-distribution data for improving the performance of the target task. In addition, [29] shows that jointly optimizing unrelated tasks leads to sparse representation, which improves performance in the target task.

Inspired by these results, we propose to use an unrelated source domain to improve performance in the target domain. To leverage and fuse the source domain's knowledge into a learning framework, we propose cross-domain mixup-noise (CDMN), which mix target samples with source samples to augment limited target labels. To effectively use the augmented target labels, we combine the CDMN with supervised contrastive learning [18]. Specifically, half of the source and half of the target samples are



(a) mini ImageNet (b) CropDisease (c) EuroSAT (d) ISIC (e) ChestX

Figure 1. Samples from source and target datasets. Classification criteria for domains are extremely different. Thereby the knowledge from one domain may not easily transfer to other domains. In particular, general knowledge will not be of much help in the medical domain, where only experts can annotate due to large intra-class and small inter-class variation.

mixed at the input level. Then, samples are passed through the shared feature extractor. Source labels supervise the source and mixed features with the source projection head; similarly, target labels supervise the target and mixed features with the target projection head. Both task losses are optimized together. In this way, not only does the model learn target-specific representations with the augmented target supervision, but it can also benefit from the regularization effect of source supervision.

To evaluate the proposed method, we modify cross-domain few-shot learning benchmark proposed by [11] shown in Figure 1 for transfer learning. Four datasets with different similarities compared to ImageNet (perspective/no perspective, natural/medical, and color/grayscale) are used with limited 1,024 images for each dataset. When our method is applied, consistent performance improvement is obtained in four scenarios with very different characteristics.

We summarize our contributions as follows:

1. We carefully design a transfer method Cross-Domain Mixup-Noise (CDMN), to leverage source domain for improving performance in the extremely different target domain.

2. We show that even though the domain gap between the source and target domains is large, augmenting the target samples using the source samples as noise helps to improve the performance.

3. We demonstrate that although the source and target domains have different label spaces, utilizing the source labels as the auxiliary task can effectively regularize and improve the target performance.

4. We achieve state-of-the-art performance in various large domain gap scenarios with a substantial margin.

5. We thoroughly analyze each component contributing to performance for transfer learning with extensive ablation studies.

2. Related Works

2.1. Transfer Learning

Transfer learning [28] aims to transfer knowledge from one domain to other domains when the amount of target data is insufficient. The standard process is to initialize the model with pretrained weights obtained from a source dataset such as ImageNet, then finetune it with a target dataset. This pretrain-finetune framework speeds up model training and substantially outperforms method training from random initialization. Several kinds of methods have been proposed to further improve the performance of finetuning.

The first is to select appropriate source samples for model training. Seq-Train [5] select top k classes of source domain which have high similarity with target domain and only use selected source classes for the pretraining stage. Selective Joint Fine-Tuning [9] selects source samples that share low-level features with target samples and jointly optimize with target samples in finetuning stage. XMixup [24] maps each target class to the most similar source class. Then, the samples of the two classes are mixed and used for finetuning.

Second, there are ways to alleviate catastrophic forgetting. Catastrophic forgetting occurs when a model loses useful features learned from pretraining as it overfits new target data. To overcome catastrophic forgetting, L2-SP [44] replaces L2 regularization to preserve source knowledge by using pretrained weights as a reference for regularization. DELTA [25] also aims to preserve source knowledge. Instead of constraining the weights, DELTA regularizes the model's output to be similar to that of the pretrained model. Although the above methods improve transfer learning performance when the domain gap between source and target domains is small, these methods rather degrade performance when the domain gap is large [3]. This paper proposes a method to achieve good performance even with extreme domain gaps.

Other works deal with negative transfer [47] that unhelpful source knowledge adversely affects target performance. BSS [3] shows that relatively small singular values of feature matrices are less transferable and suppress them to mitigate negative transfer. Several works analyze the transferability of diverse models. [31] shows adversarially trained models transfer better. [7] compares the transferability of self-supervised methods. [16] studies transfer capability of contrastive learning.

2.2. Learning from Image Mixture

In supervised learning, Mixup [46] interpolates two samples in input space and supervises them with the interpolated label to prevent memorization and learn generalizable representation. Variations such as CutMix [45] and PuzzleMix [19] have also been proposed. Manifold Mixup [35] proposes to interpolate the hidden representation instead of the input image. Several methods extend the idea of mixup to self-supervised contrastive learning. Un-Mix (Unsupervised image Mixture) [33] extends the idea of mixup to unsupervised learning to soften output and prevent over-confidence. MixSiam [10] adopts Mixup in SimSiam [2] framework and learns better representation by predicting discriminative representation using mixture of two augmented views.

Since the advent of Mixup [46], mixup has been extensively adopted with different purposes in various tasks. In domain adaptation, DM-ADA [43] uses domain mixup to learn more continuous domain invariant space. DMRL [41] exploits category and domain mixup regularizations to improve performance. CMSDA [32] mixes samples from different source domains for multi-source domain adaptation task. However, these methods assume shared label space, which hinders their widespread application. In few-shot learning literature, [8] generates mixed samples to simulate domain shift. Although this method extends the scope to different label space, few-shot learning is for learning shared representation for various few-shot episodes, which cannot be applied to the general scenario.

In this work, we apply the mixup technique in the context of the transfer learning task. Compared to the previous works that used the mixup to learn more linear representation, domain-invariant representation, or simulate domain shift, it is noteworthy that we mix samples to leverage and fuse unrelated source samples for target task learning.

3. Method

In this section, we propose a simple but strong method for improving transfer learning performance when the volume of labeled target data is limited, and the domain gap between source and target is significant. Due to the large domain gap, the knowledge learned to classify in the source domain may not be optimal for the target domain. To overcome this, we design an adaptation stage before finetuning stage to optimize the model to classify the target classes.

3.1. Adaptation Stage

In the adaptation stage, to adapt the initialized model for target classification, we propose a method augmenting limited target labels and a supervised learning framework to effectively use the augmented target labels.

3.1.1 Cross-Domain Mixup-Noise

A few studies have used cross-domain mixing to improve transfer learning performance. Given source samples, target samples, and pretrained model, XMixup [24] maps similar source class sample for each target sample and mix two samples for finetuning. In XMixup, ImageNet is used as the source domain, and datasets whose classes are related to or a subset of ImageNet (e.g., Stanford Dogs [17], or CUB-200 [38]) are used as the target domain. Since this method is based on the similarity between the source and target classes, it is difficult to apply when classes of source and target domains are unrelated.

On the other hand, we deal with the case where the similarity between the source and target domains is very low. To validate the effectiveness of our method with extreme domain gaps, we modify the benchmark from [11]. We use mini ImageNet [37], which is the subset of ImageNet, as source domain and four domains as target domains, which are CropDisease [27] (plant disease images), EuroSAT [14] (satellite images), ISIC [34, 4] (dermoscopic images), and ChestX [40] (X-ray images). As shown in Figure 1, the criterion for classification is very different between source and target datasets. In particular, in the medical domain, where the inter-class variation is very small, and the intra-class variation is very large, it is difficult for ordinary people to distinguish them, and only experts can do it. Therefore, general knowledge for classifying source samples is not optimal for classifying target samples, which limits the appli-

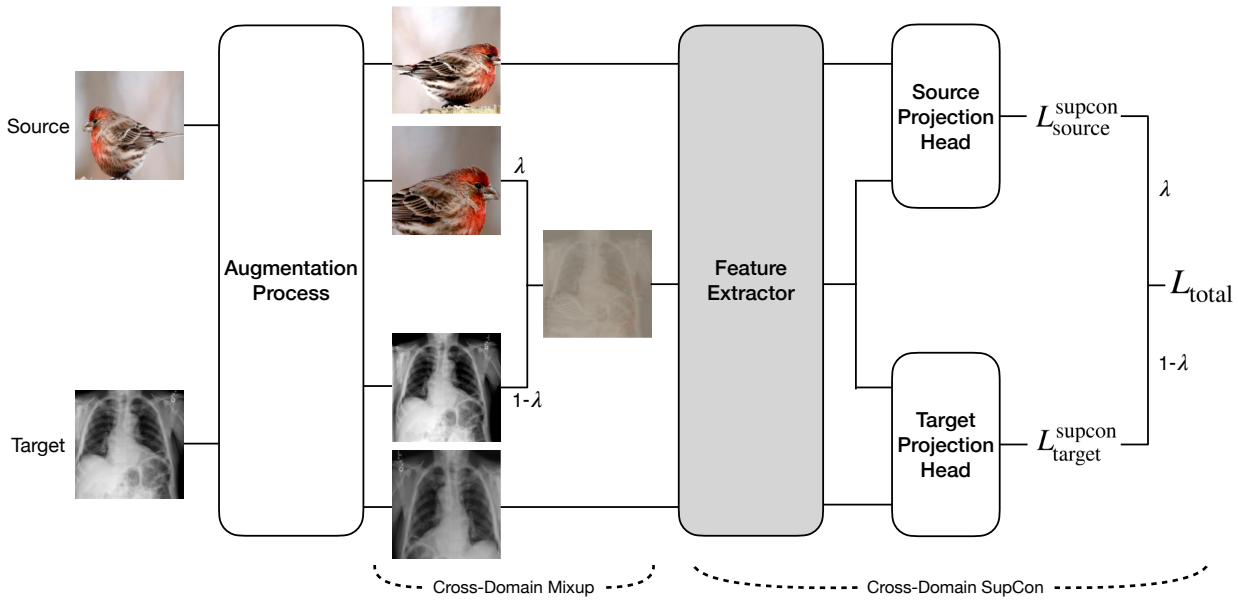


Figure 2. Method overview. **a)** Augmentation is applied to source and target samples. **b)** Half of the source and target samples are mixed with the fixed ratio, λ. **c)** Samples are forwarded through the feature extractor and the projection heads. **d)** Calculate and minimize the weighted sum of $\mathcal{L}_{\text{source}}^{\text{supcon}}$ and $\mathcal{L}_{\text{target}}^{\text{supcon}}$. In the finetuning stage, we remove the projection heads and attach a target classifier.

cation of previous methods that directly uses the similarity between two domains.

Also, since specialized domains require expensive expert knowledge for labeling, it leads to limited target training data. Data augmentation is a primary method for supplementing insufficient data. Generally, standard augmentations such as crop, rotation, color distortion, or gaussian noise are effective but still vulnerable to overfitting when the amount of data is very small. [42] points out that preferred augmentations vary according to the target task, in other words, domain-specific. (e.g., for the flower classification task, color invariance adversely affects performance, but rotation invariance helps performance). To overcome this, Domain Agnostic Contrastive Learning (DACL) [36] proposes a domain-agnostic augmentation method that mixes an anchor sample with another sample from the data manifold to generate positive samples for contrastive learning. This mixup-noise complements standard augmentations used for contrastive learning.

On the other hand, general and large-scale source samples contain diverse features which may indirectly be helpful for the target task. [23] shows that out-of-distribution(OOD) data share some features with target data, and OOD data can be used to regularize training.

Motivated by the above studies, we propose cross-domain mixup-noise (CDMN), which uses the unrelated source samples as noise to augment target training samples effectively. By randomly mixing source and target samples and supervising with target labels, we can augment expensive expert labels without reckless assumptions about domain-specific knowledge. Unlike DACL, we mix samples from two different domains and design a supervised learning framework. We find that this cross-domain mixup-noise significantly improves transfer learning performance. Notably, the proposed method does not require a similar source domain; thus, it is applicable to various scenarios.

3.1.2 Cross-Domain Joint SupCon

Along with CDMN, we use Supervised Contrastive Learning (SupCon) [18] to leverage augmented target labels effectively. SupCon extends the idea of self-supervised contrastive learning to the supervised setting. Instead of using only another augmented view from one image as a positive sample, SupCon sets samples of the same class as positive and samples from other classes as negative using the ground truth label. This prevents pushing out the features of samples belonging to the same class. Following [18], given a batch with N samples, two randomly sampled augmentations are applied to each sample, generating $2N$ samples.

For $i \in I \equiv \{1 \dots 2N\}$,

$$\mathcal{L}^{\text{supcon}} = \sum_{i \in I} \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log \frac{exp(z_i \cdot z_p / \tau)}{\sum_{a \in A(i)} exp(z_i \cdot z_a / \tau)}, \tag{1}$$

where $P(i)$ is the set of indices of samples which have the same class with i in a batch, and $A(i)$ is the set of all indices of samples except i in a batch. f is feature extractor, g is

projection head, and $z_i = g(f(i))$. τ is scaling temperature.

To effectively combine CDMN and SupCon, we propose to use source and target supervision together. Multi-task learning [1] regularizes training by jointly optimizing more than two tasks. [29, 12] even show that joint training of unrelated tasks can regularize the training of target task by preventing overfitting. We show the effect of multi-task learning with unrelated source task in the following ablation study.

The overall procedures are shown in Algorithm 1. First, we sample batch with size N from each domain and augment them to $2N$ samples using a predefined augmentation set. Then N samples from the source domain and N samples from the target domain are mixed with a specific ratio as shown in Figure 2. Then, source, target, and mixed samples are forwarded to the shared feature extractor f. Two separate projection heads (g_s and g_t) are used for the source domain and the target domain. Source projection head projects N source samples and N mixed samples, then source SupCon loss is calculated using $2N$ samples and source labels. Target SupCon loss is calculated in the same way. Then total loss is obtained, weighing two losses using the mixing ratio used in the input mixup.

$$\mathcal{L}^{\text{supcon}}_{\text{total}} = \lambda \cdot \mathcal{L}^{\text{supcon}}_{\text{source}} + (1-\lambda) \cdot \mathcal{L}^{\text{supcon}}_{\text{target}} \tag{2}$$

3.2. Finetuning Stage

After learning adapted representation in the adaptation stage, finetuning stage follows since the projection heads are not for classification. In finetuning stage, we eliminate the projection head g and add a randomly initialized classifier. When we finally finetune the model, we only update the parameters of the new classifier except for the parameters of the feature extractor. We find that overfitting occurs more easily when we finetune all parameters. We also update the feature extractor's batch normalization running statistics.

Algorithm 1 Learning algorithm of adaptation stage

Require: feature extractor f, source projection head g_s, target projection head g_t, augmentation set A, batch size N, and mixing ratio λ

for sampled minibatch $\{(x_s^k, y_s^k)\}_{k=1}^N, \{(x_t^k, y_t^k)\}_{k=1}^N$ **do**

  for all $k \in \{1, ..., N\}$ **do**

    # draw augmentations

    $a_{s1}, a_{s2}, a_{t1}, a_{t2} \sim A$

    # augment source samples

    $x_s^{2k-1} = a_{s1}(x_s^k), x_s^{2k} = a_{s2}(x_s^k)$

    # augment target samples

    $x_t^{2k-1} = a_{t1}(x_t^k), x_t^{2k} = a_{t2}(x_t^k)$

    # re-label indices and mix samples

    $x_{s'}^k = x_s^{2k-1}, x_{t'}^k = x_t^{2k-1}$

    $x_m^k = \lambda \cdot x_s^{2k} + (1-\lambda) \cdot x_t^{2k}$

    # apply feature extractor f

    $h_s^k = f(x_{s'}^k)$

    $h_t^k = f(x_{t'}^k)$

    $h_m^k = f(x_m^k)$

    # apply projection heads g_s, g_t

    $z_s^k = g_s(h_s^k), z_{ms}^k = g_s(h_m^k)$

    $z_t^k = g_t(h_t^k), z_{mt}^k = g_t(h_m^k)$

  end for

  $\mathcal{L}^{\text{supcon}}_{\text{source}} = \mathcal{L}^{\text{supcon}}(\{z_s^k, z_{ms}^k\}_{k=1}^N, \{y_s^k\}_{k=1}^N)$

  $\mathcal{L}^{\text{supcon}}_{\text{target}} = \mathcal{L}^{\text{supcon}}(\{z_t^k, z_{mt}^k\}_{k=1}^N, \{y_t^k\}_{k=1}^N)$

  $\mathcal{L}_{\text{total}} = \lambda \cdot \mathcal{L}^{\text{supcon}}_{\text{source}} + (1-\lambda) \cdot \mathcal{L}^{\text{supcon}}_{\text{target}}$

  update networks f, g_s, and g_t to minimize $\mathcal{L}_{\text{total}}$

end for

4. Experiments

4.1. Transfer Learning with Extreme Domain Gap

4.1.1 Benchmark

[11] proposed benchmark for cross-domain few-shot learning. To modify that benchmark into a transfer learning benchmark, we randomly sampled 1,024 images as a train set and the same number of images for the validation set and test set from each dataset. We used mini ImageNet [37] as the source domain. mini ImageNet is a subset of ImageNet with 64 classes with 600 images for each class. We used all samples of mini ImageNet as source training data. We used four target datasets. CropDisease [27] is plant disease image classification dataset with 38 classes. EuroSAT [14] is satellite image classification dataset with 10 classes. ISIC [34, 4] is dermoscopic image classification dataset with 7 classes. ChestX [40] is chest x-ray image classification dataset with 7 classes.

4.1.2 Implementation Details

In the adaptation stage, we followed the hyperparameter of SupCon [18]. We used a two-layer perceptron with a hidden layer with 512 dimensions and an output layer with 128 dimensions. We used SGD with a momentum of 0.9 and set the initial learning rate as 0.05. The cosine learning rate decay was applied, and the temperature was set to 0.1. We used random resized crop, image jitter, and random horizontal flip for SupCon augmentation. We set λ as 0.3 for cross-domain mixup-noise. In finetuning stage, we used cross-entropy loss, and Adam [20] optimizer with learning

Table 1. Transfer learning results with extreme domain gap.

Method	CropDisease	EuroSAT	ISIC	ChestX
Finetune Head	88.53	86.92	74.96	40.78
Finetune Head & Update BN Stat	86.50	88.32	75.66	41.27
L2	92.01	92.46	78.22	41.89
L2-SP [44]	93.95	92.71	78.24	41.55
BSS [3]	93.89	93.00	78.34	42.07
L2-SP + BSS [3]	94.00	92.56	77.83	42.03
Seq-Train [5]	91.44	92.50	77.72	42.07
Selective Joint FT [9]	83.85	79.26	70.31	40.37
XMixup [24]	85.14	84.02	74.94	33.56
StochNorm [21]	91.18	91.54	77.86	42.58
Ours	**96.19**	**94.96**	**81.07**	**44.30**

Table 2. Transfer learning results with smaller domain gap.

Method	CUB-200-2011	Stanford Cars	FGVC-Aircraft
Best Self-Supervised Method [7]	-	87.27	83.08
Robust Training [31]	-	91.48	86.24
L2	80.37	89.23	83.12
L2-SP [44]	80.04	88.63	82.51
BSS [3]	80.76	89.98	83.77
L2-SP + BSS [3]	79.36	87.38	81.27
Seq-Train [5]	78.89	87.29	80.74
Selective Joint FT [9]	80.83	88.16	82.78
XMixup [24]	81.64	90.58	84.66
StochNorm [21]	79.58	87.35	81.65
Ours	**81.69**	**92.36**	**89.80**

rate 0.001.

4.1.3 Results

Table 1 shows the results of four benchmarks. We used ResNet 10 [13] pretrained on mini ImageNet. Results are an average of 5 experiments. `Finetune Head` only updates the newly attached target classifying head, freezing the pretrained backbone's parameters, which is a standard process when target data is insufficient for training the whole model. `Finetune Head & Update BN Stat` additionally updates running statistics of batch normalization [15].

L2 finetunes all parameters of the pretrained model with L2 regularization. We set the weight for L2 regularization to 0.001. L2-SP [44], which preserves the source domain's knowledge, slightly outperformed L2 in domains other than ChestX but underperformed in ChestX, where the domain gap is extremely large. These results show that preserving source knowledge might hurt performance when the domain gap between source and target domains is large. BSS [3], which suppresses non-transferable components from source features, relatively outperformed L2-SP in target domains with larger domain gaps. Notably, BSS consistently outperformed the L2 baseline, which shows its effectiveness in extreme domain gap experiments. Combining BSS with L2-SP showed similar results.

Although retrieving source samples similar to target samples [5, 9, 24] is not straightforward when the label space is completely disjoint, and the domain gap between two domains is extremely large, we report the results of the three methods. Seq-Train [5] select top k source classes with the highest similarity with the target domain and pretrain model only with the selected source classes. Then finetune the pretrained model with the target dataset. Our implementation selects the top 32 source classes among 64 classes for pretraining. When the domain gap between source and target domains is too large, selecting the source sample degenerates to reducing the amount of data used for pretraining. Therefore, as shown in the results, Seq-Train shows similar results to L2.

Selective Joint FT [9] and XMixup [24] are finetuning-based transfer learning method. Selective Joint FT selects source images based on similarity of shallow features and jointly finetune with target data. In our implementation, we select 100 source samples for each target sample based on the similarity of the output of the first convolution block of the feature extractor. Due to the large domain gap, selected source samples have very different statistics from target samples. Results show that Selective Joint FT signifi-

cantly underperformed L2, which shows that jointly learning out-of-distribution samples in finetuning stage degrades the target performance. XMixup generates one-to-one pairing from each target class to source class using pretrained weight and finetunes the pretrained model using mixed samples. In our implementation, we set the mixing ratio λ to 0.3 for a fair comparison. Similar to Selective Joint FT, mapped source classes have very low similarity with target classes. Therefore, finetuning the pretrained model with mixed samples degrades performance compared to L2.

Our method outperformed all previous methods with a large margin. We also use source knowledge, but unlike L2-SP, which directly depends on it, we use it to regularize learning. We also use the source samples for learning. But unlike Selective Joint FT or XMixup, we did not use them for the finetuning stage but the adaptation stage. This eliminates the side effects of using out-of-distribution samples when learning decision boundaries. We analyze how each factor contributes to transfer learning performance in the subsequent ablation studies.

4.2. Transfer Learning with Smaller Domain Gap

In this paper, we focus on performing transfer learning when the domain gap is extremely large, such as natural image to medical image. In this section, we compared benchmarks with smaller domain gaps which are mainly covered in previous transfer learning studies. CUB-200-2011 [38] is the dataset for classifying bird species with 200 classes. Stanford Cars [22] is the fine-grained car dataset with 196 classes. FGVC-Aircraft [26] is the fine-grained dataset for aircraft with 102 classes.

Table 2 shows results of transfer learning from ImageNet to three target datasets with ResNet 50. [7] compares the transferability of 13 self-supervised methods, and we reported the performance of the best model. [31] compares the transferability of standard training and robust training. Results show that supervised learning (L2) outperformed self-supervised learning but underperformed robust supervised learning. Previous methods show stable performance compared to Table 1. Our method outperformed all previous methods. In particular, the larger the domain gap, the more significant the performance improvement (from CUB to Aircraft). These results show that our method is more effective when the gap between two domains is large.

5. Ablation Studies

5.1. Effect of Cross-Domain Mixup-Noise

In Table 3, we present the results of the method using various input types in the adaptation stage. `Target Only` uses vanilla SupCon loss. `Target Mixup` method generates mixed samples by mixing forward batch and reversed batch similar to [33]. `Target Mixup` showed performance improvement in several domains, demonstrating the mixup's effectiveness. In `Joint Training`, we designed to share only the feature extractor f and optimize source and target losses separately without mixup. This method showed lower performance than using only the target data, which shows using the source and target samples together does not guarantee better performance.

`Cross-Domain Mixup-Noise` showed significant performance improvement in all domains. Compared to `Target Mixup`, our method shows that jointly optimizing tasks from the different domains can improve target performance. Compared to `Joint Training`, these results show that we can get significantly different results depending on how to combine the same source data.

Table 3. Effect of cross-domain mixup-noise.

Method	CropDisease	EuroSAT	ISIC	ChestX
Target Only	94.02	93.00	79.49	42.05
Target Mixup	96.10	93.87	79.43	43.42
Joint Training	92.30	93.00	78.11	41.77
Cross-Domain Mixup-Noise	**96.19**	**94.96**	**81.07**	**44.30**

Table 4. Ablation study for mixup ratio.

Method	CropDisease	EuroSAT	ISIC	ChestX
Target Only	94.02	93.00	79.49	42.05
$\lambda = 0.1$	95.10	93.26	80.14	42.34
$\lambda = 0.2$	95.70	93.67	80.45	42.51
$\lambda = 0.3$	96.19	**94.96**	**81.07**	**44.30**
$\lambda = 0.4$	94.61	94.63	79.78	43.07
$\lambda = 0.5$	95.67	93.20	79.24	43.67
$\lambda \sim Beta(3, 7)$	**96.25**	94.41	80.22	43.93

5.2. Mixing Ratio

In Table 4, we present an ablation study for mixing ratio of `Cross-Domain Mixup-Noise` (CDMN). When the CDMN is applied, the performance is better than `Target Only` baseline in most cases. When the proportion of the source domain is too low ($\lambda = 0.1$), since the regularization effect is limited, the performance improvement is not significant. On the other hand, if the contribution of the source domain is too significant ($\lambda = 0.5$), underfitting occurs in the target domain. By mixing the source and target in an appropriate ratio ($\lambda = 0.3$), we can improve stable performance in all domains. We compared sampling λ from a beta distribution with the same mean and find fixing λ showed better results.

5.3. CE vs SupCon in Adaptation Stage

In this section, we compare the effect of supervised contrastive loss (SupCon) and cross-entropy loss (CE) in the adaptation stage. We replace two projection heads with two linear classifiers for the CE experiment. In both methods, the same class is grouped using the class label. However, the model trained on each method shows different feature representations [16].

Table 5. Comparison of CE and SupCon in adaptation stage.

Method	CropDisease	EuroSAT	ISIC	ChestX
CE	96.09	94.30	80.66	40.35
SupCon	**96.19**	**94.96**	**81.07**	**44.30**

Table 6. Effect of source supervision

Method	CropDisease	EuroSAT	ISIC	ChestX
w/o Source Supervision	95.04	91.74	79.65	43.32
w/ Source Supervision	**96.19**	**94.96**	**81.07**	**44.30**

In Table 5, we compare results from models trained on CE and SupCon. After training each loss, we attach a new classifier and only finetune it with the target data. In a relatively easy task such as CropDisease, there was no significant difference in performance between the two models. However, the performance gap enlarged as the task became more difficult. The model learns a one-to-one mapping between input and label by optimizing CE loss. Because of this characteristic, it is easy to memorize input-label pairs. On the other hand, although SupCon loss uses the same input-label pair, it creates a group based on the similarity between the two samples. We conjecture that learning the relation between samples is a more difficult to memorize and thus more robust to overfitting.

5.4. Effect of Source Supervision

In Table 6, we compare the effect of source supervision in the adaptation stage. `w/o Source Supervision` mix source and target samples in input level but removes path for $L_{\text{source}}^{\text{supcon}}$ in Figure 2. In other words, it is a method that uses only source samples as noise without using source labels for model training. `w/ Source Supervision` is our full model.

Surprisingly, joint learning with an unrelated source task significantly boosted the performance of the target task in all domains. We conjecture that this result relates to the regularization effect of multi-task learning [1]. [29] use auxiliary tasks unrelated to the target task and optimize together to get more common and sparse representation. Bayesian Multi-Domain Learning (BMDL) [12] proposes to integrate data from different domains to improve the target task's performance in the biological field. Our proposed method also combines an unrelated source task, and the results suggest that the regularization effect of the auxiliary source task can benefit transfer learning.

5.5. Comparison on How to Finetune

In Table 7, we compare three finetuning methods. `Finetune Head` freezes pretrained backbone, attaches new classifier, and trains only classifier parameters. `Finetune Head & Update BN Stat` also only trains classifier parameters, but additionally updates running mean and variance of batch normalization layer [15]. We adopted batch normalization implementation of PyTorch [30], which have learnable parameters (γ for scaling and β for shifting) and normalization statistics. The normalization statistics are tracked using statistics of training batches with exponential moving average, and the tracked normalization statistics are used in the evaluation. This method does not incur problems when the train and test data are from same distribution. However, our method used mixed samples in the adaptation stage but clean target samples in the finetuning stage, which incur distribution shifts. Therefore, normalization statistics should be updated in finetuning stage. Results show that updating batch norm statistics significantly improves performance. The results are related to the finding of Domain Specific Batch Normalization (DSBN), which uses separate batch normalization for source and target domains in domain adaptation. `Finetune All` update both parameters of the pretrained backbone and new classifier. As mentioned in the previous section, since the cross entropy loss is vulnerable to overfitting, `Finetune All` showed lower performance than `Finetune Head & Update BN Stat`.

Table 7. Comparison of finetuning method.

Method	CropDisease	EuroSAT	ISIC	ChestX
Finetune Head	95.61	94.34	80.47	41.41
Finetune Head & Update BN Stat	**96.19**	**94.96**	**81.07**	**44.30**
Finetune All	94.96	93.44	80.37	41.84

6. Conclusions

This work addressed the problem of transfer learning with extreme domain gaps. Although these scenarios are prevalent and practical, it was difficult to find studies dealing with this problem. We showed that existing methods using source knowledge did not work well when the domain gap was significant. To overcome this, we designed an adaptation stage to train a model for better target representation. We proposed cross-domain mixup-noise to augment expensive target labels using unrelated source data. We proposed a cross-domain joint supervised contrastive learning framework to leverage the augmented samples. We verified each component in the adaptation stage through extensive ablation studies and demonstrated the regularization effect of joint training of the unrelated source task. With the proposed framework, our method significantly outperformed previous methods on various benchmarks with extreme domain gaps.

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